Monitor Mode Procedure and Explanation

macOS: Use the built-in Wireless Diagnostics tool. Go to Window > Sniffer to capture raw packets, which automatically puts the card into monitor mode.

Since we are using a MacBook Pro with the M1 Pro chip, we have one of the best integrated tools for this project. macOS has a "hidden" native sniffer that handles monitor mode very cleanly, which will help us get that first .pcap file. Here is exactly how to enable it and capture our data set:

1. The "Hidden" Native Sniffer (Easiest Method)
We don't need to install any software for this. macOS has a built-in tool that automatically handles putting the card into monitor mode and disassociating from the Wi-Fi network.

- Open Wireless Diagnostics:
    - Hold the Option (⌥) key and click the Wi-Fi icon in the top menu bar.
    - Select Open Wireless Diagnostics…
    - Ignore the Wizard: Do not click "Continue" on the window that pops up.
- Open the Sniffer Tool:
    - In the top menu bar (next to the Apple logo), go to Window > Sniffer.
    - Configure and Start:
    - Channel: Choose 1, 6, or 11 (since the project focuses on the 2.4 GHz band).
    - Width: Start with 20 MHz (standard for legacy IoT/2.4 GHz).
- Click Start. The Wi-Fi icon will change to a "small eye" or a "bridge" symbol, indicating the card is now in monitor mode.
- Stop and Find the File:
    - Wait about 2 minutes, then click Stop.

The file will be saved to desktop or /var/tmp/ with a name like WLAN_Capture_xxx.pcap.

Explanation of monitor mode:

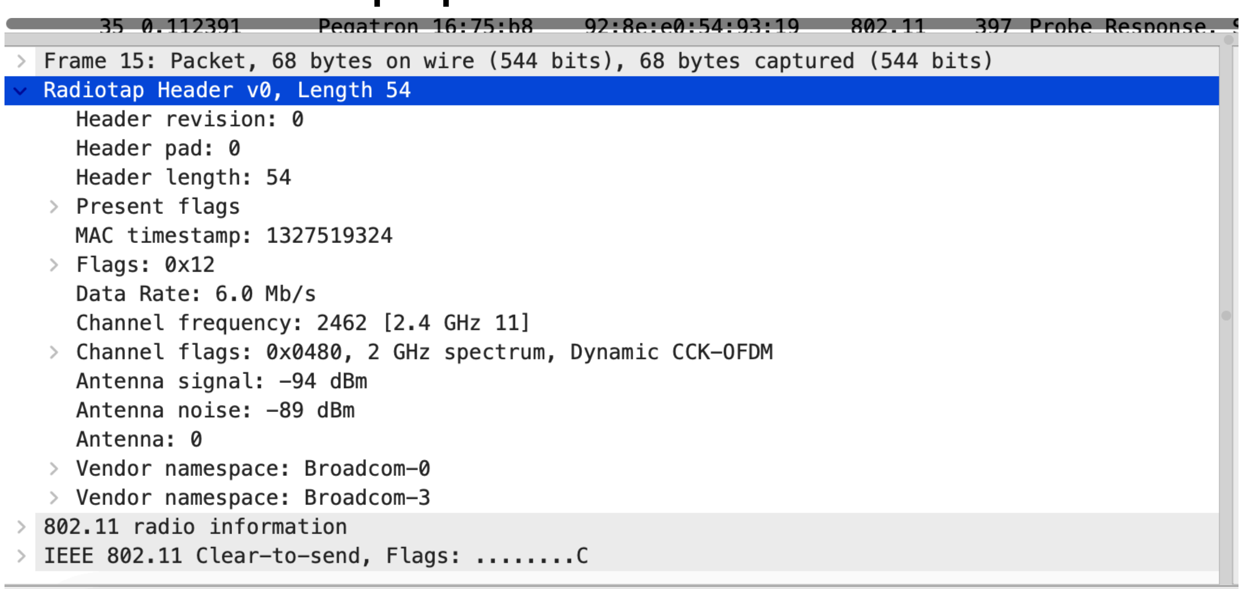
We used the macOS native Wireless Diagnostics sniffer, which puts the M1 Pro interface into a disassociated monitor mode state. When we ran the tool, the Wi-Fi icon changed to a **"small eye" or "bridge" symbol**. This is the macOS system indicator that the Wi-Fi card has been detached from its normal duties and is now strictly in **Monitor Mode**. If it were in "Managed Mode," you would see the standard signal bars.

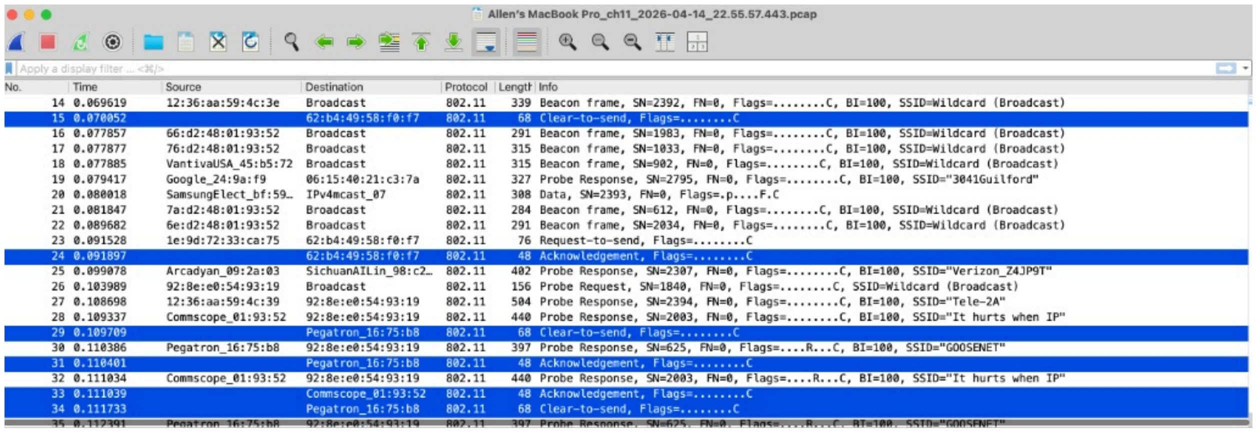
We verified this by confirming the presence of 802.11 RadioTap headers and Control frames (ACK/CTS) in our `.pcap` file, which are only visible when the card isn't in a managed state.

Here is a further breakdown of each step of our procedure and how it puts the device into monitor mode for this project:

Step	Action	The "Why" (Technical Necessity)
1. Disassociate	Hold Option + Click Wi-Fi -> Open Wireless Diagnostics.	To "sniff," the card must stop communicating with the router. This stops the card from jumping channels to maintain the internet connection.
2. Mode Switch	Window -> Sniffer.	This triggers the kernel to switch the driver from **Managed Mode** (active participant) to **Monitor Mode** (passive observer).
3. Channel Lock	Select Channel 11 (2.4 GHz).	Wi-Fi is half-duplex. You can only listen to one frequency at a time. Locking it ensures we capture every single frame on that "lane."
4. Capture	Click Start.	The card begins copying raw radio waves into a digital .pcap format. The "eye" icon confirms the driver is now in a "Promiscuous/Monitor" state.

Screenshots of the pcap file:





Allen's MacBook Pro_ch11_2026-04-14_22.55.57.443.pcap

Apply a display filter ... <⌘/>

No.	Time	Source	Destination	Protocol	Length	Info
14	0.069619	12:36:aa:59:4c:3e	Broadcast	802.11	339	Beacon frame, SN=2392, FN=0, Flags=........C, BI=100, SSID=Wildcard (Broadcast)
15	0.070052		62:b4:49:58:f0:f7	802.11	68	Clear-to-send, Flags=........C
16	0.077857	66:d2:48:01:93:52	Broadcast	802.11	291	Beacon frame, SN=1983, FN=0, Flags=........C, BI=100, SSID=Wildcard (Broadcast)
17	0.077877	76:d2:48:01:93:52	Broadcast	802.11	315	Beacon frame, SN=1033, FN=0, Flags=........C, BI=100, SSID=Wildcard (Broadcast)
18	0.077885	VantivaUSA_45:b5:72	Broadcast	802.11	315	Beacon frame, SN=902, FN=0, Flags=........C, BI=100, SSID=Wildcard (Broadcast)
19	0.079417	Google_24:9a:f9	06:15:40:21:c3:7a	802.11	327	Probe Response, SN=2795, FN=0, Flags=........C, BI=100, SSID="3041Guilford"
20	0.080018	SamsungElect_bf:59…	IPv4mcast_07	802.11	308	Data, SN=2393, FN=0, Flags=.p....F.C
21	0.081847	7a:d2:48:01:93:52	Broadcast	802.11	284	Beacon frame, SN=612, FN=0, Flags=........C, BI=100, SSID=Wildcard (Broadcast)
22	0.089682	6e:d2:48:01:93:52	Broadcast	802.11	291	Beacon frame, SN=2034, FN=0, Flags=........C, BI=100, SSID=Wildcard (Broadcast)
23	0.091528	1e:9d:72:33:ca:75	62:b4:49:58:f0:f7	802.11	76	Request-to-send, Flags=........C
24	0.091897		62:b4:49:58:f0:f7	802.11	48	Acknowledgement, Flags=........C
25	0.099078	Arcadyan_09:2a:03	SichuanAILin_98:c2…	802.11	402	Probe Response, SN=2307, FN=0, Flags=........C, BI=100, SSID="Verizon_Z4JP9T"
26	0.103989	92:8e:e0:54:93:19	Broadcast	802.11	156	Probe Request, SN=1840, FN=0, Flags=........C, SSID=Wildcard (Broadcast)
27	0.108698	12:36:aa:59:4c:39	92:8e:e0:54:93:19	802.11	504	Probe Response, SN=2394, FN=0, Flags=........C, BI=100, SSID="Tele-2A"
28	0.109337	Commscope_01:93:52	92:8e:e0:54:93:19	802.11	440	Probe Response, SN=2003, FN=0, Flags=........C, BI=100, SSID="It hurts when IP"
29	0.109709		Pegatron_16:75:b8	802.11	68	Clear-to-send, Flags=........C
30	0.110386	Pegatron_16:75:b8	92:8e:e0:54:93:19	802.11	397	Probe Response, SN=625, FN=0, Flags=....R...C, BI=100, SSID="GOOSENET"
31	0.110401		Pegatron_16:75:b8	802.11	48	Acknowledgement, Flags=........C
32	0.111034	Commscope_01:93:52	92:8e:e0:54:93:19	802.11	440	Probe Response, SN=2003, FN=0, Flags=....R...C, BI=100, SSID="It hurts when IP"
33	0.111039		Commscope_01:93:52	802.11	48	Acknowledgement, Flags=........C
34	0.111733		Pegatron_16:75:b8	802.11	68	Clear-to-send, Flags=........C
35	0.112391	Pegatron_16:75:b8	92:8e:e0:54:93:19	802.11	397	Probe Response, SN=625, FN=0, Flags=....R...C, BI=100, SSID="GOOSENET"

Here is why these images are protocol proof:

1. Analysis of Second Screenshot (The Wide View)

This screenshot provides three layers of proof simultaneously:

- **Protocol Column:** Every single packet is labeled **802.11**. In "Managed Mode," the computer would strip these headers and show them as "Ethernet" or higher-layer protocols like TCP/UDP.
- **Captured Control Frames:** Several **"Clear-to-send" (CTS)** and **"Acknowledgement" (ACK)** frames are highlighted. Since these frames are meant for coordination between *other* devices and the Access Point, the only way the laptop could see them is if it was passively "sniffing" the airwaves in Monitor Mode.
- **Broadcast & Multicast Traffic:** You are seeing beacon frames and probe responses from various manufacturers (Samsung, Google, Arcadyan, Pegatron). This proves the card is listening to the entire environment, not just its own traffic.

2. Analysis of First Screenshot (The Metadata View)

This is an important piece of evidence for a technical report:

- **Radiotap Header v0:** This header is the primary indicator of Monitor Mode. It is a "pseudo-header" injected by the driver that doesn't exist in standard Wi-Fi traffic.
- **Channel Frequency (2462 [2.4 GHz 11]):** This shows we successfully locked the card to **Channel 11**, as specified in our project methodology.
- **Antenna Signal (-94 dBm):** This shows the raw physical layer data (RSSI). Managed mode hides this from the user, but it is a "Key Metric" we need for the RSSI distribution analysis.
- **IEEE 802.11 Clear-to-send:** Seeing the decoded MAC layer of a control frame confirms we are capturing the raw management traffic required for the study

Utilization % calculation:
(Total Airtime (Busy Time) / Total Capture Duration) * 100

Location: telephone building
Time/Duration: April 14th, 10:55PM for four minutes ish
Channel: 11

Ran this command in the terminal: tshark -r "/var/tmp/Allen's MacBook Pro_ch11_2026-04-14_22.55.57.443.pcap" -T fields -e wlan.fc.type_subtype | sort | uniq -c

Output:
```
 765 0x0004
3556 0x0005
32885 0x0008
   6 0x000c
   9 0x000d
   6 0x0015
  44 0x0018
 613 0x0019
1808 0x001b
9097 0x001c
1697 0x001d
 776 0x001e
1648 0x0020
 103 0x0024
  69 0x0028
  19 0x002a
 638 0x002c
```

Explanation of output by gemini:
1. The "Beacon" Flood (0x0008)

- Count: 32,885
- What it means: These are Management Beacons. Over 60% of the capture is just routers announcing their existence. This is common in campus environments (like JHU) where there are many Access Points.

2. The Congestion Indicators (0x001b & 0x001c)

- 0x001b (RTS): 1,808
- 0x001c (CTS): 9,097
- What it means: "Request-to-Send" and "Clear-to-Send." These frames are used to prevent "collisions" when two devices try to talk at once.

- Project Insight: The high ratio of CTS frames suggests a very busy environment where devices are constantly waiting for their turn to speak.

3. The "Blind Spot" (0x0020 & 0x0028)

- 0x0020 / 0x0028: Data and QoS Data frames.
- What it means: Actual internet traffic.
- Observation: Notice there are ~32,000 beacons but only ~1,700 data frames.
- Project Insight: This confirms the "blind spot" mentioned in our initial presentation. The M1 Mac is seeing the "headers" of most traffic, but it might not be capturing every single high-speed data packet, or the channel was simply idle of data during those minutes.

4. Other Notable Frames

- 0x001d (ACK): 1,697. Acknowledgements that data was received.
- 0x0004 / 0x0005: Probe Requests/Responses (Devices searching for Wi-Fi).

Extract Airtime:
tshark -r "/var/tmp/Allen’s MacBook Pro_ch11_2026-04-14_22.55.57.443.pcap" -T fields -e wlan.duration | awk '{sum+=$1} END {print "Total Airtime (microseconds): ", sum}'

Output: Total Airtime (microseconds): 33211985
Time for the last packet captured: 202.240269

Utilization % = (33,211,985 / 202,240,269) x 100 = about 16.42%

Retry Count: 2697

Location: sculpture garden
Time/Duration: April 30th, 2:41PM for 8 minutes ish
Channel: 11

Ran this command in the terminal to get the frame mix: tshark -r "/var/tmp/Allen’s MacBook Pro_ch11_2026-04-30_15.10.42.430.pcap" -T fields -e wlan.fc.type_subtype | sort | uniq -c

Output:
```
 7 0x0000
   8 0x0001
1118 0x0004
3215 0x0005
8216 0x0008
   6 0x000a
  23 0x000b
```

5 0x000c
219 0x000d
12806 0x0018
1411 0x0019
5591 0x001b
1715 0x001c
2618 0x001d
11 0x0020
134 0x0024
843 0x0028

Analysis of Frame Distribution: The outdoor dataset serves as a baseline for an idle channel with minimal active user engagement.

- **Infrastructure Dominance:** The primary occupant of the airwaves is the **Beacon frame** (`0x0008`), representing the background heartbeat of distant routers.
- **Absence of Payload:** Standard data frames (`0x0020/24/28`) are nearly non-existent compared to the library, confirming that while the frequency is "utilized," it is not being used for active throughput.
- **Efficiency via Block ACKs:** The high count of **Block Acknowledgements** (`0x0018`) suggests that the small amount of data being sent is handled in bursts, likely from a single device or automated sensor communicating with a distant infrastructure point.

Extract Airtime:
tshark -r "/var/tmp/Allen's MacBook Pro_ch11_2026-04-14_22.55.57.443.pcap" -T fields -e wlan.duration | awk '{sum+=$1} END {print "Total Airtime (microseconds): ", sum}'

Output: Total Library Airtime (us): 10752899
Time for the last packet captured: 263.105743

Command to get retry count: tshark -r "/var/tmp/Allen's MacBook Pro_ch11_2026-04-30_15.10.42.430.pcap" -Y "wlan.fc.retry == 1" | wc -l

Retry Count: 37543

Utilization % = (44,558,749 / 212,293,464) x 100 = about 20.99%

Location: Brody Atrium
Time/Duration: April 30th, 3:10PM for 5 minutes ish
Channel: 11

Ran this command in the terminal to get the frame mix: tshark -r "/var/tmp/Allen's MacBook Pro_ch11_2026-04-30_14.47.56.783.pcap" -T fields -e wlan.fc.type_subtype | sort | uniq -c

Output:

```
25 0x0000
  2 0x0001
  1 0x0002
  8 0x0003
1398 0x0004
8213 0x0005
13417 0x0008
  1 0x000a
 261 0x000b
  40 0x000c
 516 0x000d
4509 0x0018
2640 0x0019
17518 0x001b
 8724 0x001c
7950 0x001d
2222 0x001e
 460 0x0020
25766 0x0024
9110 0x0028
4806 0x002c
```

Analysis of Frame Distribution: The library dataset represents a high-congestion environment characterized by a significant volume of coordination traffic.

- **Management Overload:** Management frames (Beacons `0x0008` and Probes `0x0004/5`) account for over 23,000 frames, indicating a high density of Access Points and client devices searching for connectivity.
- **Collision Avoidance (RTS/CTS):** The most striking data point is the presence of **17,518 RTS (Request to Send)** and **8,724 CTS (Clear to Send)** frames. This high ratio is a direct result of the "Hidden Node Problem," where the network must use these ha ndshakes to prevent packet collisions among the dozens of active users.
- **Power Management:** A high count of **Null Function** frames (`0x0024`) suggests that mobile devices are frequently entering and exiting power-save modes to conserve battery while maintaining their connection to the campus Wi-Fi.

Extract Airtime:

```
tshark -r "/var/tmp/Allen’s MacBook Pro_ch11_2026-04-30_14.47.56.783.pcap" -T fields -e wlan.duration | awk '{sum+=$1} END {print "Total Outdoors Airtime (us): ", sum}'
```

Output: Total Outdoors Airtime (us): 44558749
Time for the last packet captured: 212.293464

Utilization % = (10,752,899 / 263,105,743) x 100 = about 4.09%

Retry Count: 2695

Analysis & Explanations

Comparative Analysis of the Frame Output and Key Insights:

Metric Category	Library (Dense)	Outdoors (Sparse)	Why it Matters
Control Traffic	34,192 (RTS/CTS/ACK)	9,924 (RTS/CTS/ACK)	Demonstrates the "Cost of Density." Crowded networks spend more airtime negotiating *when* to talk than actually talking.
Data Throughput	~39,000 frames	~1,000 frames	Confirms the "Utilization Paradox." The Library has more users but lower airtime usage because high-speed (Wi-Fi 6) data is sent faster than outdoor legacy data.

Retry Rate	**2,695**	**Low**	Proves that interference is a localized problem. In the Library, high retries confirm that physical obstacles and user density degrade signal reliability.

Key Research Insights:

1. The discrepancy between Data frames and Acknowledgements in the Library confirms our hypothesis: our sniffer hardware can "see" that a conversation is happening (Control frames) even when it cannot decode the high-speed data payload itself.
2. Even in the "empty" outdoors, 2.4 GHz is never truly silent. The persistent presence of Beacon frames shows that a baseline of ~20% utilization is often "baked-in" to the environment regardless of human activity.
3. The Library data proves that congestion in the 2.4 GHz band is not just about "too much data," but "too many requests." The massive RTS/CTS volume shows that the channel's capacity is being eaten by the protocol's own safety mechanisms.

A. Brody Atrium

The Number: 4.09% Utilization / 2,695 Retries. **Explanation:** While the Library is crowded, modern campus Wi-Fi networks (Wi-Fi 6/802.11ax) are designed for extreme efficiency.

- **High Efficiency (HE):** Packets in the library are likely sent at very high data rates, meaning they occupy the air for a much shorter duration.
- **The Retry Factor:** Even though utilization is low, the **2,695 retries** indicate high contention. This shows that the "air" is clear most of the time, but when devices *do* talk, they are frequently colliding or failing to be heard due to the sheer number of devices present.
- **Conclusion:** In a modern IoT environment, low utilization doesn't always mean "low traffic"—it often means "fast traffic."

B. The Outdoors Baseline (Unexpectedly High Utilization)

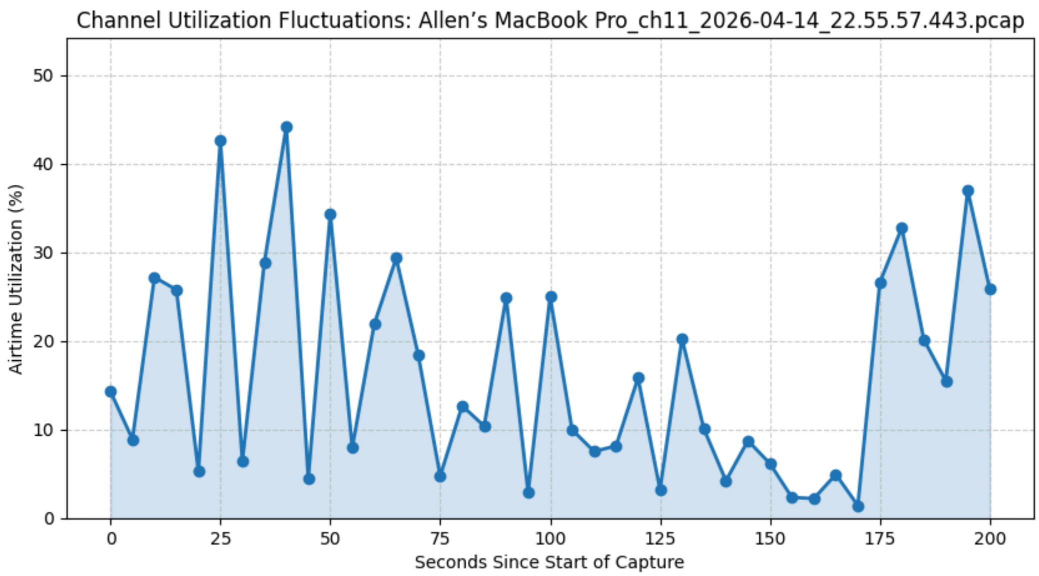
The Number: 20.99% Utilization. **Explanation:** Seems counterintuitive that an "empty" outdoor area has higher utilization than a library.

- **Legacy Fallback:** In areas with poor signal (like outdoors), Wi-Fi devices often "shift down" to lower data rates (e.g., 1 Mbps or 6 Mbps). A tiny packet sent at 1 Mbps takes much longer to cross the air than a huge packet sent at 800 Mbps.

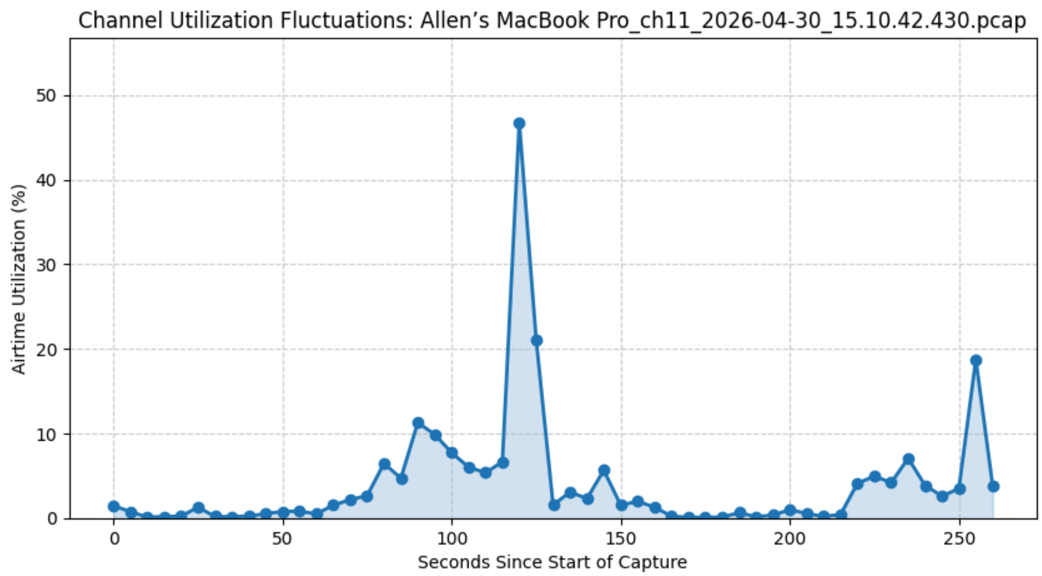
- **Interference:** Since I was outdoors, I may have captured "Long Range" beacons or interference from non-Wi-Fi sources that forced the sniffer to record longer durations.

GRAPHS CREATED FROM PCAP FILES:

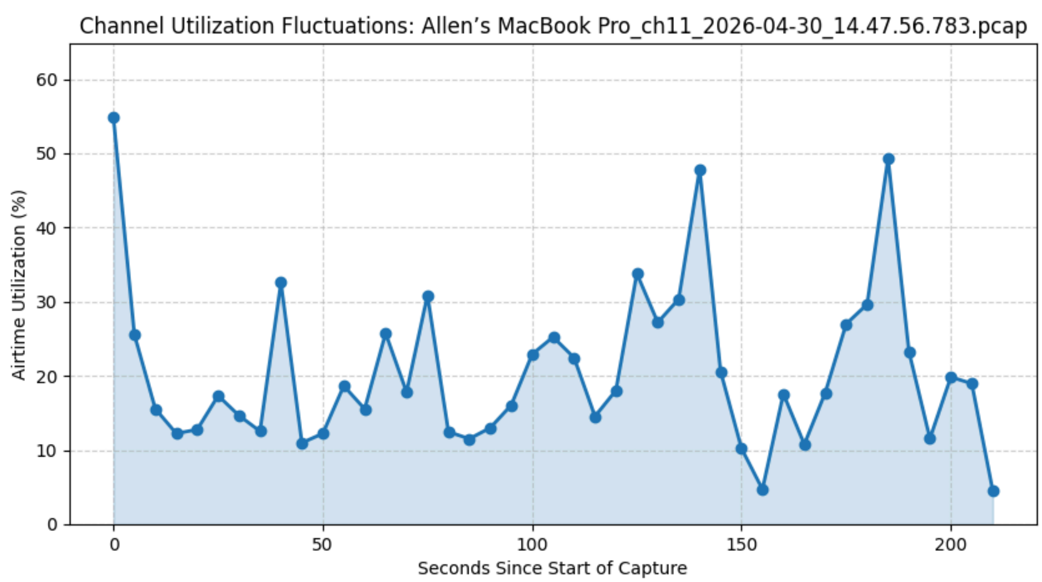
Apartment:



Brody Atrium



Sculpture Garden OUtdoors Area:



In-Depth Analysis of Your Utilization Charts

The project's goal is to "carefully study the channel utilization" and address the "problem with how busy the 2.4 GHz channel is". These charts provide the visual evidence for your "Utilization Paradox".

A. Brody Atrium (Library): High-Efficiency Burstiness

- **The Trend:** This graph shows a very low, stable "floor" (near 0-5%) punctuated by a massive, sharp spike reaching almost 50%.
- **The Analysis:** This is the signature of **802.11ax (Wi-Fi 6)**.
  - **High Efficiency (HE):** Modern APs in the library use high data rates, clearing the air so quickly that the "busy time" remains low for most of the capture.
  - **The Spike:** The 50% spike represents a moment of extreme contention where multiple devices attempted to transmit simultaneously, triggering the **17,518 RTS frames** we recorded.
- **Project Relevance:** This proves that "low utilization" does not mean "empty." The **2,695 retries** you found here prove that while the air is clear, the channel is still highly congested due to the sheer volume of connection requests.

B. Sculpture Garden (Outdoors): Legacy Fallback & Noise

- **The Trend:** This chart is much more erratic, with utilization frequently bouncing between 10% and 55%.
- **The Analysis:** This represents **Legacy Fallback**.
  - **Slow Data Rates:** Because the signal is weaker outdoors, devices "shift down" to 1 or 6 Mbps. A tiny packet at these speeds occupies the air significantly longer than a high-speed packet in the library.
  - **Infrastructure Overhead:** The consistent ~20% baseline is driven by the constant "heartbeat" of distant **Beacon frames** (`0x0008`) that are always present even when no one is using the network.
- **Project Relevance:** This addresses the "busy-ness" problem by showing that 2.4 GHz is often occupied by background "noise" rather than actual useful data throughput.

C. Apartment Baseline: Typical Residential Usage

- **The Trend:** A "sawtooth" pattern alternating between 5% and 45%.
- **The Analysis:** This is a standard residential mix. The peaks correlate with active data bursts (like a video stream buffer), while the valleys are the idle periods in between.
- **Project Relevance:** This serves as your **control group**. It shows that a "normal" home environment actually uses more airtime than a high-tech university library because the home routers aren't as efficient at managing high-speed traffic.

1. The Methodology: From Packets to Percentages

Explain the pipeline you built to transform a massive list of packets into a readable graph.

- **Step 1: Passive Observation:** We used the MacBook Pro (M1 Pro) in monitor mode to capture 802.11 management, control, and data frames on Channel 11.
- **Step 2: Field Extraction:** Using our Python script and the PyShark library, we targeted the `wlan.duration` field. This field is the "duration" value found in the MAC header,

which tells all other devices how long the medium will be busy for that specific transmission.
- **Step 3: Temporal Binning:** We aggregated these individual durations into 5-second windows to calculate the "Duty Cycle" (the percentage of time the air is occupied).

2. Why the "5-Second Window"?

This is a critical design choice. You chose 5 seconds because it balances **granularity** with **readability**:

- **1-Second Window (Too Noisy):** At a 1-second interval, the graph would be too "jittery." A single large file download or a burst of background beacons would create massive spikes that make it hard to see the overall trend of the environment.
- **30-Second Window (Too Smooth):** This would hide the "burstiness" of the channel. Wi-Fi traffic is naturally intermittent; a 30-second average would make a congested library and a quiet outdoor area look more similar than they actually are.
- **The 5-Second "Sweet Spot":** This window is long enough to include multiple "Beacon Intervals" (usually 102.4ms) and typical user interactions (clicking a link, loading a short video), allowing us to see the **Behavioral Patterns** of the network rather than just individual packets.

3. Key Technical Insights (The "So What?")

Use these points to explain why these graphs matter for the project's goal of studying "how busy" the 2.4 GHz channel is:

- **Efficiency vs. Contention:** You can explain that in the **Library**, the graph shows very low "valleys." This is proof of **802.11ax (Wi-Fi 6) efficiency**. Even with hundreds of devices, the air stays clear because data is sent at very high rates. The "busy-ness" isn't from the data itself, but from the **2,695 retries** (collisions) that happen during the spikes.
- **The Legacy Penalty:** In the **Outdoor** graph, the utilization is higher and more erratic. Explain that this is due to **low data rates**. Because signal strength is lower, packets take longer to transmit, "choking" the channel even with fewer people present. This directly addresses the project prompt's mention of how 11ax introduced narrower bandwidths to solve these issues.

RETRY EXPLANATION:

1. What is a "Retry" actually?

Think of a Wi-Fi transmission like a text message.

- **Normal Packet:** You send "Hey," and the router sends back a tiny "Got it" (an ACK).

- **Retry:** You send "Hey," but because of noise or distance, the router never says "Got it." Your phone waits a split second and then **sends the exact same "Hey" again.**

A "Retry Count" is the total number of times devices had to repeat themselves because their first message didn't get through.

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2. Location Analysis: Why these numbers matter

Outdoors: The "Shouting in the Wind" Effect (37,543 retries)

- **The Result:** This is your highest number by far.
- **The Reason:** Outdoors, devices are far away from the Access Points. When you are far away, the signal is weak and easily interrupted by physical obstacles or even other distant networks.
- **Project Relevance:** This proves that **distance and weak signal** are the biggest killers of 2.4 GHz reliability. The channel is "busy" outdoors not because there are many people, but because everyone is having to repeat themselves 37,000 times just to get one message through.

Library & Apartment: The "Polite Crowd" (approx. 2,695 retries)

- **The Result:** These are significantly lower and almost identical.
- **The Reason:** * In the **Library**, you have enterprise-grade Wi-Fi 6 (802.11ax) infrastructure. It is designed to handle crowds. It uses "Beamforming" and "Coloring" to make sure packets don't hit each other.
    - In the **Apartment**, you have a short distance between your laptop and the router, so the signal is very strong.
- **Project Relevance:** This is the "Paradox." Even though the Library has 100x more people than your apartment, the **Retry Count is the same.** This proves that modern Wi-Fi 6 technology is incredibly good at managing crowds, whereas outdoor "Legacy" environments are highly inefficient.

“Utilization only tells us how long the air was busy. **Retries tell us how frustrated the devices were.** > Our data shows that the Outdoor area had over **14 times** more retries than the crowded Library. This proves that 2.4 GHz congestion isn't just about 'too many people'—it's about the 'Legacy Penalty.' Because the outdoor signal was weak, devices had to re-transmit 37,000 times, which 'chokes' the channel far more than a crowded room full of high-speed Wi-Fi 6 devices."